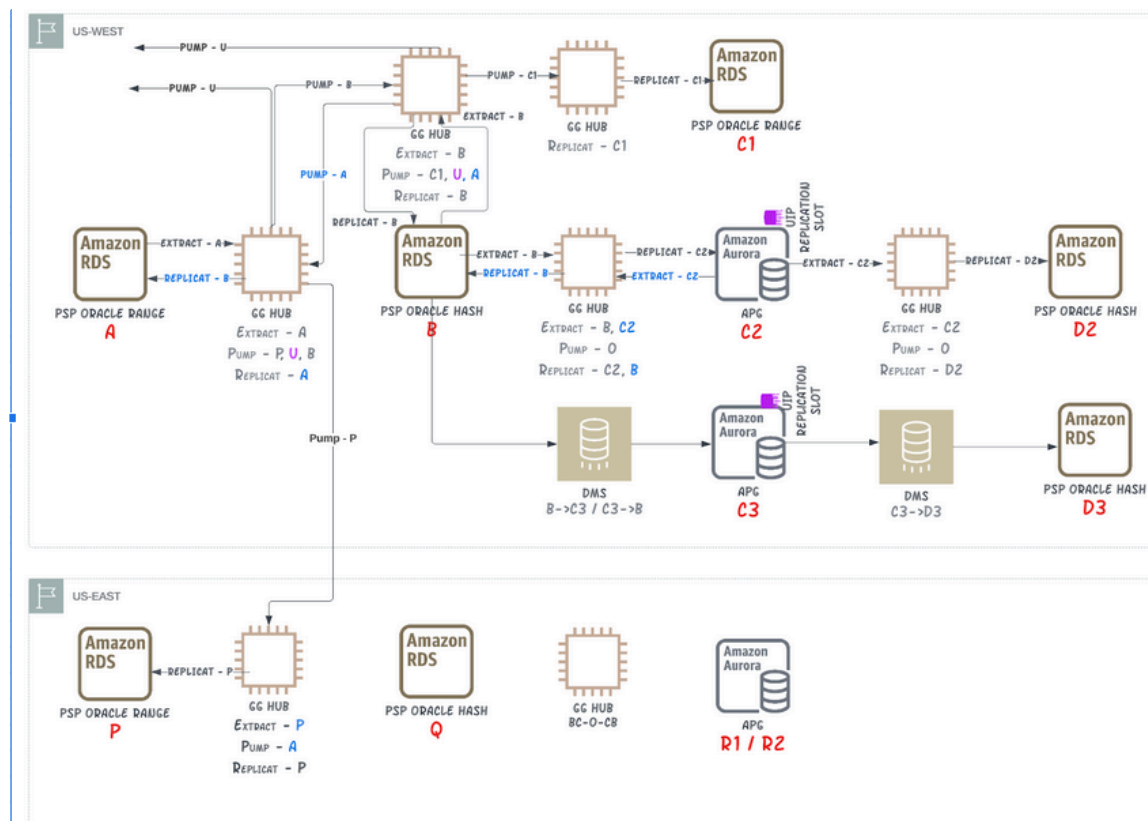


PSP Prod Monolith Aurora PostgreSQL migration

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Replication Configuration



West	PSP Monolith Prod	Instance Name	East	PSP Monolith Prod
A	Oracle (Range)	pspuwp01	P	Oracle (Range)
B	Oracle (HASH)	psphpp02	Q	Oracle (HASH)
B(Replica)	Oracle(HASH)	psphdg02	R1	APG (HASH) GG
C1	Oracle(Range)	psprpp01	R2	APG (HASH) DMS
C1'	Oracle(Range)	psprpp02	S1	Oracle (HASH) GG
C2	APG (HASH) GG	psp-prod-uw02,pspapg02	S2	Oracle (HASH) DMS
C3	APG (HASH) DMS	psp-prod-mon,psppp01		
D2	Oracle (HASH) GG - source C2	psphpp05		
D3	Oracle (HASH) DMS - source C3	psphpp06		
D2'	Oracle (HASH) DMS - source C2	psphpp07		

Current database object layout

Monolith

	OWNER	OBJECT_TYPE	OBJECT_COUNT
1	PSPADM	DATABASE LINK	3
2	PSPADM	FUNCTION	13
3	PSPADM	INDEX	834
4	PSPADM	INDEX PARTITION	3758
5	PSPADM	INDEX SUBPARTITION	4830
6	PSPADM	LOB	30
7	PSPADM	LOB PARTITION	464
8	PSPADM	LOB SUBPARTITION	3220
9	PSPADM	PACKAGE	3
10	PSPADM	PACKAGE BODY	3

1 1	PSPADM	PROCEDURE	28
1 2	PSPADM	SEQUENCE	76
1 3	PSPADM	SYNONYM	2
1 4	PSPADM	TABLE	383
1 5	PSPADM	TABLE PARTITION	1931
1 6	PSPADM	TABLE SUBPARTITION	805
1 7	PSPADM	TRIGGER	24
1 8	PSPADM	VIEW	5

	TABLE_NAME	No. Of Partitions
1	PSP_ENTITY_UPDATE	4
2	PSP_ENTRY_DETAIL_RECORD	187
3	PSP_FINANCIAL_TRANSACTION	94
4	PSP_FINANCIAL_TRANS_STATE	187
5	PSP_LEDGER_BALANCE	33
6	PSP_MONEY_MOVEMENT_TRANSACTION	94
7	PSP_PAYCHECK	32
8	PSP_PAYCHECK_SPLIT	32

Assessment

Summary

Monolith

Figure: Conversion statistics for database storage objects

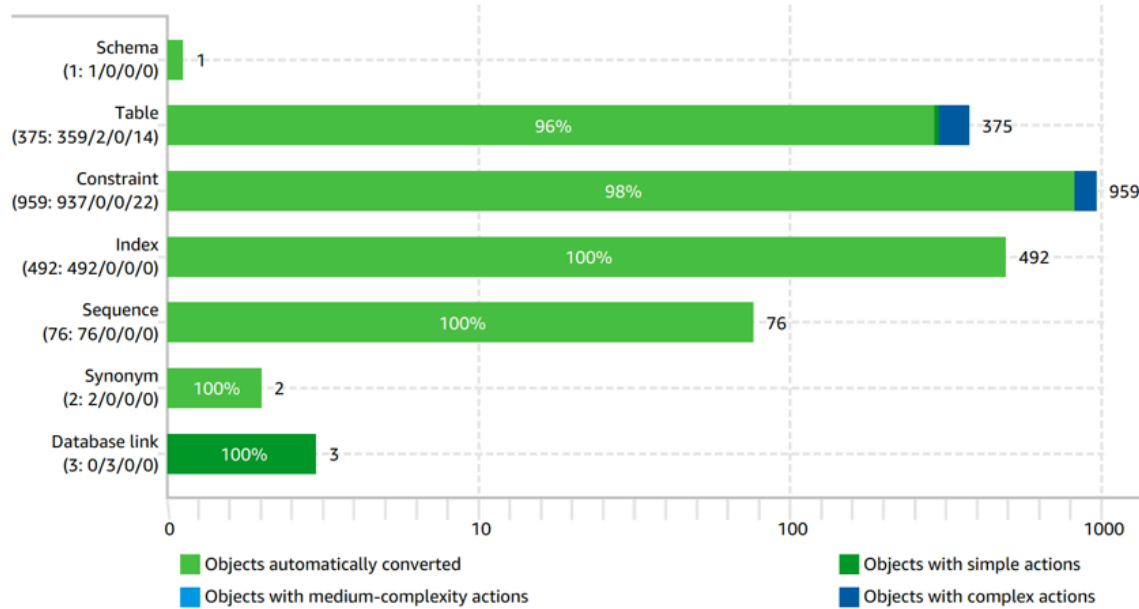
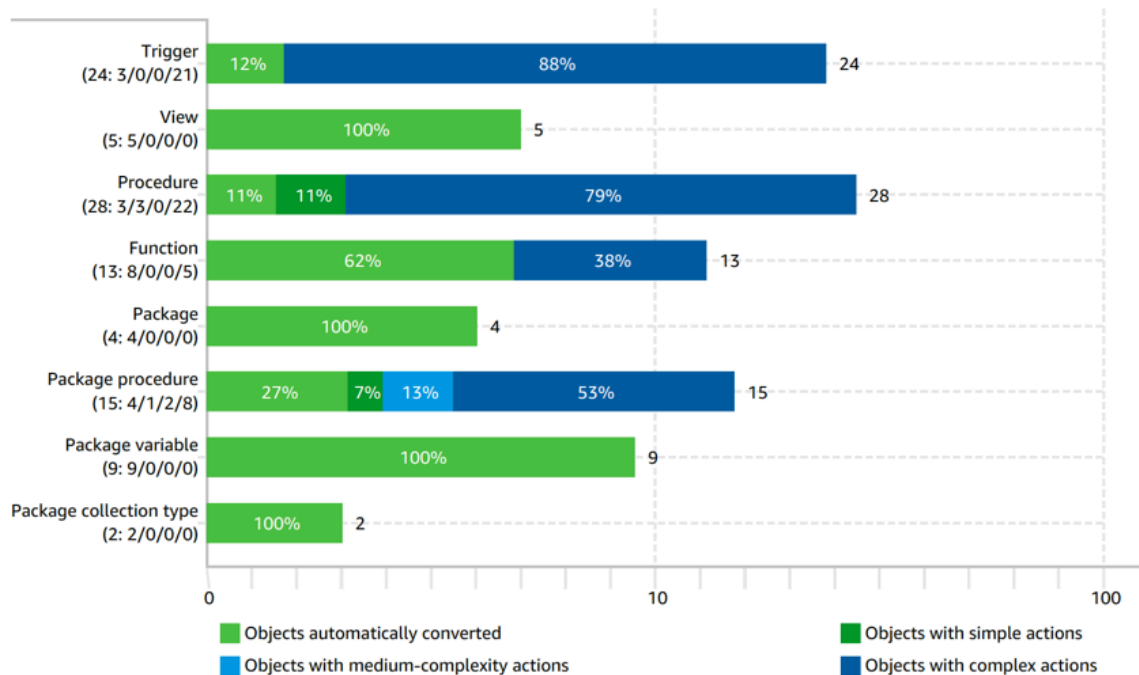


Figure: Conversion statistics for database code objects



No statistics available for database code objects for IBOB/Secondary database.

Full Report

[psp_prod_monolith_sct_migration_assessment_report_aurora_postgres.pdf](#)

Approach

- Migration would be two step approach
 - Schema refactoring on Oracle
 - Migrate to PostgreSQL
- Refactoring includes removal of **relam_id** column from PK for all tables and adding **company_fk** as PK to all partitioned tables and its childs.

Schema/Code Changes

- Primary Key
 - Remove REALM_ID and Add company_fk to PK for all partitioned tables.
 - This is to required to have hash partitioning based on company_fk.
- Partitioning
 - (Why Composite Primary Key?) This is prerequisite to partition by company_fk in Postgres. [Reference](#)
- Data Integrity
 - If Company_fk column exists then Populate company_fk column in all child tables otherwise add company_fk column in all child tables & populate it.
- Indexes
 - All Primary Key & Foreign Key indexes needs to be changed to reflect Primary Key change.
 - All other Indexes needs to be scrubbed to exclude REALM_ID.
- Performance
 - Ensure queries come to database with company_fk filter

	Database Objects	Change in Oracle Hash	Change in Postgres Hash
1	Table	• Global removal of RELAM_ID reference • PK change to composite key • Tablespace distribution of tables & Indexes	• Global removal of RELAM_ID reference • PK change to composite key
2	Indexes (Secondary & FK indexes)	• All Partitioned table PK & FK indexes changed to reflect primary key changes • Removed REALM_ID references from all indexes	• All Partitioned table PK & FK indexes changed to reflect primary key changes • Removed REALM_ID references from all indexes
3	FK Constraints	• All Partitioned table PK & FK constraints changed to reflect primary key changes • Removed REALM_ID references from all FK constraints	• All Partitioned table PK & FK constraints changed to reflect primary key changes • Removed REALM_ID references from all FK constraints

4 Tables without PK

```

1 SCHEMA_NAME, TABLE_NAME
2 PSPADM, ARCH_PURGE_MAIN_STATS_LI
  ST
3 PSPADM, AS400_DROPME
4 PSPADM, BACKUP_SCHEMA_STATS_0909
  11
5 PSPADM, BACKUP_STATS
6 PSPADM, BACKUP_STATS_032911
7 PSPADM, BACKUP_TAB_STATS_090911
8 PSPADM, CMP3$100488
9 PSPADM, CMP3$15077
10 PSPADM, CMP4$15077
11 PSPADM, DEL_TEST
12 PSPADM, DISTINCT_COMPANY_FK
13 PSPADM, EMPTY
14 PSPADM, PAYCHECK_BACKUP_DATE
15 PSPADM, PAYROLL_HIST
16 PSPADM, PET_FEB27
17 PSPADM, PSP_COMPANY_EVENT_BKP
18 PSPADM, PSP_COMPANY_NOTE_BKP
19 PSPADM, PSP_EMPLOYEE_BANK_ACCOUN
  T_BKP
20 PSPADM, PSP_FINANCIAL_TRANS_STAT
  E_BKP
21 PSPADM, PSP_LB_TMP
22 PSPADM, PSP_LB_TMP_COMP_FK
23 PSPADM, PSP_LB_TMP_COMP_FK_5000
24 PSPADM, PSP_LEDGER_BALANCE_BKP
25 PSPADM, PSP_POSTING_RULE_MAR7
26 PSPADM, PSP_PROPERTY_AUDIT_BKP
27 PSPADM, PSP_RPT_PAID_EMPLOYEES
28 PSPADM, PSP_SAP_METHOD_CALL_BKP
29 PSPADM, PSP_TRANSACTION_TYPE_MAR
  7
30 PSPADM, REP_TEST
31 PSPADM, STATS_11DEC
32 PSPADM, TEMP1_PAYCHECK
33 PSPADM, TEMP_A_PRC_INC
34 PSPADM, TEMP_BAD_PAYROLL
35 PSPADM, TEMP_BKP_EDR_JUNE3RD
36 PSPADM, TEMP_COMPANY_07152010
37 PSPADM, TEMP_COMPANY_DATAFILE
38 PSPADM, TEMP_DD4V_PRC_INC
39 PSPADM, TEMP_DD_PRC_INC
40 PSPADM, TEMP_ENTITLE_PENDING
41 PSPADM, TEMP_FIX_NCD
42 PSPADM, TEMP_FIX_NCD_REACTIVATIO
  N
43 PSPADM, TEMP_FT
44 PSPADM, TEMP_FUTURE_PAYROLL_DATA
45 PSPADM, TEMP_GUID_COMPANY
46 PSPADM, TEMP_MODES_AGENCY_IDS
47 PSPADM, TEMP_ORSTT_AGENCY_IDS
48 PSPADM, TEMP_PAYCHECK
49 PSPADM, TEMP_PEDR_FEB23
50 PSPADM, TEMP_PFTS_FEB23
51 PSPADM, TEMP_PFT_FEB23
52 PSPADM, TEMP_PMMT_FEB23
53 PSPADM, TEMP_PSP_PAYROLL_FRAUD_B
  ATCH
54 PSPADM, TEMP_RAFFI_SQL_LOG
55 PSPADM, TEMP_SUBSCRIPTIONNUMBER
56 PSPADM, TEMP_TAX_B
57 PSPADM, TEMP_TAX_C
58 PSPADM, TEMP_TAX_CALC_APR12
59 PSPADM, TEMP_TAX_D
60 PSPADM, TEMP_TAX_E
61 PSPADM, TEST2
62 PSPADM, TEST_MMT_ATF
63 PSPADM, TMP_AUTH_USER
64 PSPADM, TMP_AUTH_USER1
65 PSPADM, TMP_PEW_FEB27
66 PSPADM, TMP_PFT_FEB26
67 PSPADM, TMP_PMMT_FEB25
68 PSPADM, TMP_PSP_EDR
69 PSPADM, TMP_PSP_MMT
70 PSPADM, TMP_PTTMD_FEB27
71 PSPADM, TOAD_PLAN_SQL
72 PSPADM, T_BACKFILL_METADATA
73 PSPADM, T_PCEEP_OLD
74 PSPADM, T_PDATL_OLD
75 PSPADM, T_PPPI_OLD
76 PSPADM, T_PPU

```

		77 PSPADM,T_PPU_OLD 78 PSPADM,Z_REDEF_PAYCHECK 79 PSPADM,Z_REDEF_PAYCHECK_SPLIT 80 PSPADM,Z_REDEF_SRC_SYS_TRANSMI SION	
5	Tables which are not migrated/skipped	PSP_ENTITY_UPDATE_HIST	
6			

Detailed Schema Refactor checks

- [PSP Schema Refactor Checks](#)

Manual Validations Performed

- Total object types and its count
- Invalid Objects
- Total Tables
- Partitioned Tables and its partition count
- All Primary Key Columns
- All Primary Key Indexes
- All Index types and its count
- All constraint types and its count'
Compare All CHECK and NOT NULL COSNSTRAINTS from production
- Enabled Triggers Count
- Partitioned PK indexes and its partition count
- Local indexes and its partition count
- All FK constraint Names, FK constraint columns and respective Primary table, primary table columns
- Only FK indexes'

Schema Comparison via DB solo or liquibase

```

1  --PSP_FINANCIAL_TRANSACTION
2  create table PSPADM.PSP_FINANCIAL_TRANSACTION
3  (
4      FINANCIAL_TRANSACTION_SEQ VARCHAR2(255) not null
5      constraint SYS_C008669
6      check ("FINANCIAL_TRANSACTION_SEQ" IS NOT NULL),
7      VERSION NUMBER(19) not null
8      constraint SYS_C008670
9      check ("VERSION" IS NOT NULL),
10     CREATOR_ID VARCHAR2(30),
11     CREATED_DATE TIMESTAMP(6) not null
12     constraint SYS_C008671
13     check ("CREATED_DATE" IS NOT NULL),
14     MODIFIER_ID VARCHAR2(30),
15     MODIFIED_DATE TIMESTAMP(6) not null
16     constraint SYS_C008672
17     check ("MODIFIED_DATE" IS NOT NULL),
18     REALM_ID NUMBER(19) default -1 not null
19     constraint SYS_C008673
20     check ("REALM_ID" IS NOT NULL),
21     FINANCIAL_TRANSACTION_AMOUNT NUMBER(19,4),
22     SETTLEMENT_DATE TIMESTAMP(6),
23     SETTLEMENT_TYPE_CD VARCHAR2(255)
24     constraint C_PSP_FINANCIAL_TRANSACTION0
        check (SETTLEMENT_TYPE_CD IN('EFE', 'ACH', 'Wire',
        'Cash', 'CheckType', 'Other', 'ApplyForward', 'HPDE', 'TakenOnReturn', 'EFTPS', 'EFTPSDirectDebit', 'EDI')),

```

```

25 CREDIT_BANK_ACCOUNT_TYPE VARCHAR2(255)
26     constraint C_PSP_FINANCIAL_TRANSACTION1 check (CREDIT_BANK_ACCOUNT_TYPE IN('Company',
'Employee', 'Intuit', 'TaxAgency', 'Payee')),
27 DEBIT_BANK_ACCOUNT_TYPE VARCHAR2(255)
28     constraint C_PSP_FINANCIAL_TRANSACTION2 check (DEBIT_BANK_ACCOUNT_TYPE IN('Company', 'Employee',
'Intuit', 'TaxAgency', 'Payee')),
29 ON_HOLD NUMBER(1),
30 SKU VARCHAR2(40),
31 SKU_QUANTITY NUMBER(10),
32 BILLING_DETAIL_FK VARCHAR2(255),
33 CREDIT_BANK_ACCOUNT_FK VARCHAR2(255),
34 DEBIT_BANK_ACCOUNT_FK VARCHAR2(255),
35 COMPANY_FK VARCHAR2(255),
36 PAYROLL_RUN_FK VARCHAR2(255),
37 PAYCHECK_SPLIT_FK VARCHAR2(255),
38 TRANSACTION_TYPE_FK VARCHAR2(255) not null
39     constraint SYS_C008674 check ("TRANSACTION_TYPE_FK" IS NOT NULL),
40 LAW_FK VARCHAR2(255),
41 CURRENT_TRANSACTION_STATE_FK VARCHAR2(255),
42 ORIGINAL_TRANSACTION_FK VARCHAR2(255),
43 MONEY_MOVEMENT_TRANSACTION_FK VARCHAR2(255),
44 ORIGINAL_SETTLEMENT_DATE TIMESTAMP(6),
45 REFUND_TYPE VARCHAR2(255 char)
46     constraint C_PSP_FINANCIAL_TRANSACTION3 check (REFUND_TYPE IN('ApplyForward', 'Immediate',
'TOR')),
47 COMP_ADJUST_SUBMISSION_FK VARCHAR2(255 char),
48 TAX_PENALTY_INTEREST_FK VARCHAR2(255 char),
49 RELATABLE_TRANSACTION_FK VARCHAR2(255 char),
50 BILL_PAYMENT_SPLIT_FK VARCHAR2(255 char),
51 COMPANY_LAW_FK VARCHAR2(255 char),
52     constraint SYS_C008675
53     primary key (FINANCIAL_TRANSACTION_SEQ, REALM_ID), -- Remove REALM_ID & add COMPANY_FK. NEW PK
(COMPANY_FK, FINANCIAL_TRANSACTION_SEQ)
54     constraint PSP_FINANCIAL_TRANSACTION_FK1 foreign key (BILLING_DETAIL_FK, REALM_ID) references
PSPADM.PSP_BILLING_DETAIL, -- Remove REALM_ID
55     constraint PSP_FINANCIAL_TRANSACTION_FK2 foreign key (RELATABLE_TRANSACTION_FK, REALM_ID) references
PSPADM.PSP_FINANCIAL_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the same
56     constraint PSP_FINANCIAL_TRANSACTION_FK3 foreign key (ORIGINAL_TRANSACTION_FK, REALM_ID) references
PSPADM.PSP_FINANCIAL_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the same
57     constraint PSP_FINANCIAL_TRANSACTION_FK10 foreign key (MONEY_MOVEMENT_TRANSACTION_FK, REALM_ID) references
PSPADM.PSP_MONEY_MOVEMENT_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the
same
58     constraint PSP_FINANCIAL_TRANSACTION_FK11 foreign key (COMP_ADJUST_SUBMISSION_FK, REALM_ID) references
PSPADM.PSP_COMP_ADJUST_SUBMISSION, -- Remove REALM_ID
59     constraint PSP_FINANCIAL_TRANSACTION_FK13 foreign key (BILL_PAYMENT_SPLIT_FK, REALM_ID) references
PSPADM.PSP_BILL_PAYMENT_SPLIT, -- Remove REALM_ID
60     constraint PSP_FINANCIAL_TRANSACTION_FK14 foreign key (TAX_PENALTY_INTEREST_FK, REALM_ID) references
PSPADM.PSP_TAX_PENALTY_INTEREST, -- Remove REALM_ID
61     constraint PSP_FINANCIAL_TRANSACTION_FK16 foreign key (COMPANY_LAW_FK, REALM_ID) references
PSPADM.PSP_COMPANY_LAW, -- Remove REALM_ID
62     constraint PSP_FINANCIAL_TRANSACTION_FK2 foreign key (CREDIT_BANK_ACCOUNT_FK, REALM_ID) references
PSPADM.PSP_BANK_ACCOUNT, -- Remove REALM_ID
63     constraint PSP_FINANCIAL_TRANSACTION_FK3 foreign key (DEBIT_BANK_ACCOUNT_FK, REALM_ID) references
PSPADM.PSP_BANK_ACCOUNT, -- Remove REALM_ID
64     constraint PSP_FINANCIAL_TRANSACTION_FK4 foreign key (COMPANY_FK, REALM_ID) references PSPADM.PSP_COMPANY, --
Remove REALM_ID
65     constraint PSP_FINANCIAL_TRANSACTION_FK5 foreign key (PAYROLL_RUN_FK, REALM_ID) references
PSPADM.PSP_PAYROLL_RUN, -- Remove REALM_ID
66     constraint PSP_FINANCIAL_TRANSACTION_FK6 foreign key (PAYCHECK_SPLIT_FK, REALM_ID) references
PSPADM.PSP_PAYCHECK_SPLIT, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the same
67     constraint PSP_FINANCIAL_TRANSACTION_FK7 foreign key (TRANSACTION_TYPE_FK, REALM_ID) references
PSPADM.PSP_TRANSACTION_TYPE, -- Remove REALM_ID
68     constraint PSP_FINANCIAL_TRANSACTION_FK8 foreign key (LAW_FK, REALM_ID) references PSPADM.PSP_LAW, --
Remove REALM_ID
69     constraint PSP_FINANCIAL_TRANSACTION_FK9 foreign key (CURRENT_TRANSACTION_STATE_FK, REALM_ID) references
PSPADM.PSP_TRANSACTION_STATE -- Remove REALM_ID )
70 /
71 ----- Child table -----
72 --PSP_QBDT_TRANSACTION_INFO
73 create table PSPADM.PSP_QBDT_TRANSACTION_INFO
74 (
75     QBDT_TRANSACTION_INFO_SEQ VARCHAR2(255 char) not null,
76     VERSION NUMBER(19) not null,
77     CREATOR_ID VARCHAR2(30 char),
78     CREATED_DATE TIMESTAMP(6) not null,
79     MODIFIER_ID VARCHAR2(30 char),
80     MODIFIED_DATE TIMESTAMP(6) not null,
81     REALM_ID NUMBER(19) default -1 not null,
82     AGENCY_NAME VARCHAR2(128 char),
83     REFERENCE_NUMBER VARCHAR2(11 char),
84     ACCOUNT_NAME VARCHAR2(128 char),
85     MEMO VARCHAR2(4000 char),

```



```

86     ON_SERVICE NUMBER(1),
87     CLEARED VARCHAR2(1 char),
88     TRACKING_CLASS VARCHAR2(128 char),
89     IS_DELETED NUMBER(1),
90     TOKEN NUMBER(19) default -2,
91     LIABILITY_CHECK_FK VARCHAR2(255 char),
92     LIABILITY_CHECK_LINE_FK VARCHAR2(255 char),
93     MONEY_MOVEMENT_TRANSACTION_FK VARCHAR2(255 char),
94     IS_DIRECT_DEPOSIT NUMBER(1) default 0,
95     SYSTEM_GENERATED NUMBER(1) default 0,
96     COMP_ADJUST_SUBMISSION_FK VARCHAR2(255 char),
97     LIABILITY_ADJUSTMENT_FK VARCHAR2(255 char),
98     FINANCIAL_TRANSACTION_FK VARCHAR2(255 char),
99     PRIOR_PAYMENT_SUBMISSION_FK VARCHAR2(255 char),
100    COMPANY_FK VARCHAR2(255 char),
101    QBDT_PAYROLL_TRANSACTION_FK VARCHAR2(255 char),
102    QBDT_PAYROLL_TRANS_LINE_FK VARCHAR2(255 char),
103    primary key (QBDT_TRANSACTION_INFO_SEQ, REALM_ID), -- remove relam_id from PK
104    constraint PSP_QBDT_TRANSACTION_INFO_FK1 foreign key (PRIOR_PAYMENT_SUBMISSION_FK, REALM_ID) references
PSPADM.PSP_PRIOR_PAYMENT_SUBMISSION,
105    constraint PSP_QBDT_TRANSACTION_INFO_FK1 foreign key (LIABILITY_CHECK_FK, REALM_ID) references
PSPADM.PSP_LIABILITY_CHECK,
106    constraint PSP_QBDT_TRANSACTION_INFO_FK2 foreign key (LIABILITY_CHECK_LINE_FK, REALM_ID) references
PSPADM.PSP_LIABILITY_CHECK_LINE,
107    constraint PSP_QBDT_TRANSACTION_INFO_FK3 foreign key (MONEY_MOVEMENT_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_MONEY_MOVEMENT_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK to reflect
the same
108    constraint PSP_QBDT_TRANSACTION_INFO_FK4 foreign key (FINANCIAL_TRANSACTION_FK, REALM_ID) references
PSPADM.PSP_FINANCIAL_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the same
109    constraint PSP_QBDT_TRANSACTION_INFO_FK5 foreign key (COMP_ADJUST_SUBMISSION_FK, REALM_ID) references
PSPADM.PSP_COMP_ADJUST_SUBMISSION,
110    constraint PSP_QBDT_TRANSACTION_INFO_FK6 foreign key (LIABILITY_ADJUSTMENT_FK, REALM_ID) references
PSPADM.PSP_LIABILITY_ADJUSTMENT,
111    constraint PSP_QBDT_TRANSACTION_INFO_FK7 foreign key (COMPANY_FK, REALM_ID) references
PSPADM.PSP_COMPANY,
112    constraint PSP_QBDT_TRANSACTION_INFO_FK8 foreign key (QBDT_PAYROLL_TRANSACTION_FK, REALM_ID)
references PSPADM.PSP_QBDT_PAYROLL_TRANSACTION,
113    constraint PSP_QBDT_TRANSACTION_INFO_FK9 foreign key (QBDT_PAYROLL_TRANS_LINE_FK, REALM_ID)
references PSPADM.PSP_QBDT_PAYROLL_TRANS_LINE
114 )
115 /
116
117 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK1 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(PRIOR_PAYMENT_SUBMISSION_FK, REALM_ID); --remove REALM_ID
118 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK1 on PSPADM.PSP_QBDT_TRANSACTION_INFO (LIABILITY_CHECK_FK,
REALM_ID); --remove REALM_ID
119 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK2 on PSPADM.PSP_QBDT_TRANSACTION_INFO (LIABILITY_CHECK_LINE_FK,
REALM_ID); --remove REALM_ID
120 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK3 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(MONEY_MOVEMENT_TRANSACTION_FK, REALM_ID); --remove REALM_ID and add COMPANY_FK to reflect FK. Eg
(COMPANY_FK, MONEY_MOVEMENT_TRANSACTION_FK)
121 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK4 on PSPADM.PSP_QBDT_TRANSACTION_INFO (FINANCIAL_TRANSACTION_FK,
REALM_ID); --remove REALM_ID and add COMPANY_FK to reflect FK. Eg (COMPANY_FK, FINANCIAL_TRANSACTION_FK)
122 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK5 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(COMP_ADJUST_SUBMISSION_FK, REALM_ID); --remove REALM_ID
123 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK6 on PSPADM.PSP_QBDT_TRANSACTION_INFO (LIABILITY_ADJUSTMENT_FK,
REALM_ID); --remove REALM_ID
124 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK7 on PSPADM.PSP_QBDT_TRANSACTION_INFO (COMPANY_FK, REALM_ID); --
remove REALM_ID
125 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK8 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(QBDT_PAYROLL_TRANSACTION_FK, REALM_ID); --remove REALM_ID
126 create index PSPADM.PSP_QBDT_TRANSACTION_INFO_FK9 on PSPADM.PSP_QBDT_TRANSACTION_INFO
(QBDT_PAYROLL_TRANS_LINE_FK, REALM_ID); --remove REALM_ID
127
128 --PSP_TRANSACTION_OFFLOAD_BATCH
129 create table PSPADM.PSP_TRANSACTION_OFFLOAD_BATCH
130 (
131     TRANSACTION_OFFLOAD_BATCH_SEQ VARCHAR2(255 char) not null,
132     VERSION NUMBER(19) not null,
133     CREATOR_ID VARCHAR2(30 char),
134     CREATED_DATE TIMESTAMP(6) not null,
135     MODIFIER_ID VARCHAR2(30 char),
136     MODIFIED_DATE TIMESTAMP(6) not null,
137     REALM_ID NUMBER(19) default -1 not null,
138     OFFLOAD_STATUS_CD VARCHAR2(10 char),
139     STATUS_EFFECTIVE_DATE TIMESTAMP(6),
140     FINANCIAL_TRANSACTION_FK VARCHAR2(255 char) not null,
141     OFFLOAD_BATCH_FK VARCHAR2(255 char) not null,
142     primary key (TRANSACTION_OFFLOAD_BATCH_SEQ, REALM_ID), --Remove REALM_ID
143     constraint PSP_TRANSACTION_OFFLOAD_BATCH_FK1 foreign key (FINANCIAL_TRANSACTION_FK, REALM_ID) references
PSPADM.PSP_FINANCIAL_TRANSACTION, -- PK is changing removing REALM_ID and COMPANY_FK to reflect the same. Eg
(COMPANY_FK, FINANCIAL_TRANSACTION_FK)

```

```

144  constraint PSP_TRANSACTION_OFFLOAD_BA_FK2 foreign key (OFFLOAD_BATCH_FK, REALM_ID) references
    PSPADM.PSP_OFFLOAD_BATCH --remove REALM_ID
145  )
146  /
147
148  create index PSPADM.PSP_TRANSACTION_OFFLOAD_BA_FK2 on PSPADM.PSP_TRANSACTION_OFFLOAD_BATCH (OFFLOAD_BATCH_FK,
    REALM_ID); --remove REALM_ID
149  create index PSPADM.PSP_TRANSACTION_OFFLOAD_BA_FK1 on PSPADM.PSP_TRANSACTION_OFFLOAD_BATCH
    (FINANCIAL_TRANSACTION_FK, REALM_ID); --remove REALM_ID and add COMPANY_FK to reflect FK. Eg
    (COMPANY_FK, FINANCIAL_TRANSACTION_FK)

```

Partitioning Principles

- Consider table for partitioning whose size is >100G. This is to handle organic & inorganic growth for tables.
- Partition tables by tenant id (company_fk). This will also help in horizontal scalability of database.
- Consider tables sizes and growth rate (estimated/projected next 3 year and 5 year)
- Keep same number of partitions for logical grouping of tables as per workflows
- Keep individual partition size less than or equal to 50G
- Distribute tables across multiple tablespaces to ensure we do not hit AWS RDS big data file size limit.
- Review partitioned tables for OLTP & OLAP traffic to confirm on number of partitions
- partitioned tables should have company_fk filter

Query Analysis POC

- **Partitioned Table Join With Non Partitioned Table**

Filter/Join	Queries	Query Plan	Observations
Both + company_fk join	<pre> 1 explain select * 2 from pspadmhashdms.psp_money _movement_transaction mmt, 3 pspadmhashdms.psp_atfp ayments_to_process atp 4 where mmt.money_movement_tran saction_seq = atp.money_movement_tran saction_fk 5 and mmt.COMPANY_FK ='81ea198c-6fa4-4a19- abcd-705c93605e43' 6 and atp.COMPANY_FK ='81ea198c-6fa4-4a19- abcd-705c93605e43' 7 ; </pre> <pre> 1 explain select * 2 from pspadmhashdms.psp_money _movement_transaction mmt, 3 pspadmhashdms.psp_atfp ayments_to_process atp 4 where mmt.money_movement_tran saction_seq = atp.money_movement_tran saction_fk 5 and mmt.company_fk = atp.company_fk 6 and mmt.COMPANY_FK ='81ea198c-6fa4-4a19- abcd-705c93605e43' 7 and atp.COMPANY_FK ='81ea198c-6fa4-4a19- </pre>	<pre> 1 QUERY PLAN 2 ----- 3 Merge Join (cost=1.39..3967.18 rows=2 width=1352) 4 Merge Cond: ((mtt.money_movement_tr ansaction_seq)::text = (atp.money_movement_tra nsaction_fk)::text) 5 -> Index Scan using psp_money_movement_tran saction_hash_p3_pkey on psp_money_movement_tran saction_p3 mmt (cost=0.70..2589.48 rows=2502 width=1153) 6 Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19- abcd- 705c93605e43'::text) 7 -> Index Scan using psp_atfpayments_to_proc ess_fk1 on psp_atfpayments_to_proc ess atp (cost=0.69..1368.09 rows=1337 width=199) 8 Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19- abcd- 705c93605e43'::text) </pre>	Company_fk join is not required.

	<pre> 8 abcd-705c93605e43' ; </pre>	<pre> 9 (6 rows) </pre>	
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• Partitioned Table Join With Partitioned Table

Filter/Join	Queries	Query Plan	Observations
Both + company_fk join	<pre> 1 select * 2 from pspadmhashdms.psp_money _movement_transaction mmt, 3 pspadmhashdms.psp_finan cial_transaction ft 4 where mmt.money_movement_tran saction_seq = ft.money_movement_trans action_fk 5 and mmt.COMPANY_FK = ft.COMPANY_FK 6 and mmt.COMPANY_FK = '81ea198c-6fa4-4a19- abcd-705c93605e43' 7 and ft.COMPANY_FK = '81ea198c-6fa4-4a19- abcd-705c93605e43'; </pre> <pre> 1 select * 2 from pspadmhashdms.psp_money _movement_transaction mmt, 3 pspadmhashdms.psp_finan cial_transaction ft 4 where mmt.money_movement_tran saction_seq = ft.money_movement_trans action_fk 5 and mmt.COMPANY_FK = '81ea198c-6fa4-4a19- abcd-705c93605e43' 6 and ft.COMPANY_FK = '81ea198c-6fa4-4a19- abcd-705c93605e43'; </pre>	same plan for both queries <pre> 1 2 QUERY PLAN 3 4 ----- 5 6 ----- 7 8 ----- 9 10 ----- 11 12 Merge Join (cost=1.39..4806.82 rows=22756 width=3289) 13 Merge Cond: ((ft.money_movement_tr ansaction_fk)::text = (mtt.money_movement_tr ansaction_seq)::text) 14 -> Index Scan using psp_financial_transact ion_fk10_p3 on psp_financial_transact ion_p3 ft (cost=0.69..1871.46 rows=1819 width=2136) 15 16 Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19- abcd- 705c93605e43'::text) 17 -> Materialize (cost=0.70..2595.73 rows=2502 width=1153) 18 -> Index Scan using psp_money_movement_tra nsaction_hash_p3_pkey on psp_money_movement_tra nsaction_p3 mmt (cost=0.70..2589.48 rows=2502 width=1153) 19 20 Index Cond: ((company_fk)::text = '81ea198c-6fa4-4a19- abcd- 705c93605e43'::text) 21 (7 rows) </pre>	Company_fk join is not required.

• 2 Partitioned Tables Join With Non-Partitioned Table

2 Partit	<pre> 1 explain select * 2 from pspadmhashdms.psp_money_move </pre>	<pre> 1 QUERY PLAN </pre>	Compa ny_fk

ional /Non - partit ion table s filters /join	<pre> 3 ment_transaction mmt, 4 pspadmhashdms.psp_financial_ transaction ft, 5 pspadmhashdms.psp_atfpayment s_to_process atp 6 where mmt.money_movement_transacti on_seq = ft.money_movement_transactio n_fk 7 and mmt.money_movement_transacti on_seq = atp.money_movement_transacti on_fk 8 and mmt.COMPANY_FK = ft.COMPANY_FK 9 and mmt.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43' 10 and ft.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43' 11 and atp.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43'; </pre> <pre> 1 explain select * 2 from pspadmhashdms.psp_money_movem ent_transaction mmt, 3 pspadmhashdms.psp_financial_t ransaction ft, 4 pspadmhashdms.psp_atfpayments _to_process atp 5 where mmt.money_movement_transactio n_seq = ft.money_movement_transaction _fk 6 and mmt.money_movement_transactio n_seq = atp.money_movement_transactio n_fk 7 and mmt.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43' 8 and ft.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43' 9 and atp.COMPANY_FK = '81ea198c-6fa4-4a19-abcd- 705c93605e43'; </pre>	<pre> 2 ----- 3 ----- 4 -- 5 Nested Loop (cost=2.08..3250.17 rows=18 width=3488) 6 -> Merge Join (cost=1.38..3247.45 rows=1 width=2335) 7 Merge Cond: 8 ((ft.money_movement_transaction_fk)::text = 9 (atp.money_movement_transaction_fk)::text) 10 -> Index Scan using psp_financial_transaction_fk10_p3 on psp_financial_transaction_p3 ft (cost=0.69..1871.46 rows=1819 width=2136) 11 Index Cond: ((company_fk)::text = '81ea198c- 6fa4-4a19-abcd-705c93605e43'::text) 12 -> Index Scan using psp_atfpayments_to_process_fk1 on psp_atfpayments_to_process atp (cost=0.69..1368.09 rows=1337 width=199) 13 Index Cond: ((company_fk)::text = '81ea198c- 6fa4-4a19-abcd-705c93605e43'::text) 14 -> Index Scan using psp_money_movement_transaction_hash_p3_pkey on psp_money_movement_transaction_p3 mmt (cost=0.70..2.71 rows=1 width=1153) 15 Index Cond: (((company_fk)::text = '81ea198c-6fa4- 4a19-abcd-705c93605e43'::text) AND ((money_movement_transaction_seq)::text = (ft.money_movement_transaction_fk)::text)) (9 rows) </pre>	join is not required

Schema Conversion

Data Type Mapping

Criteria followed for target datatype w.r.t source NUMER datatype

Precision(m)	Scale(n)	Oracle	PostgreSQL	Comments
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<5	0	NUMBER(m,n)	SMALLINT	Documentation Page
5 >= m <= 9	0	NUMBER(m,n)	INT	
10<= m <=18	0	NUMBER(m,n)	BIGINT	
m+n <= 15	n>0	NUMBER(m,n)	DOUBLE PRECISION	
m+n > 15	n>0	NUMBER(m,n)	NUMERIC	

NUMBER(19,4) and NUMBER(19,7) Decision

- Use NUMERIC with same precision and scale as that of Oracle

NUMBER(10,0) and NUMBER(19,0) Optimization

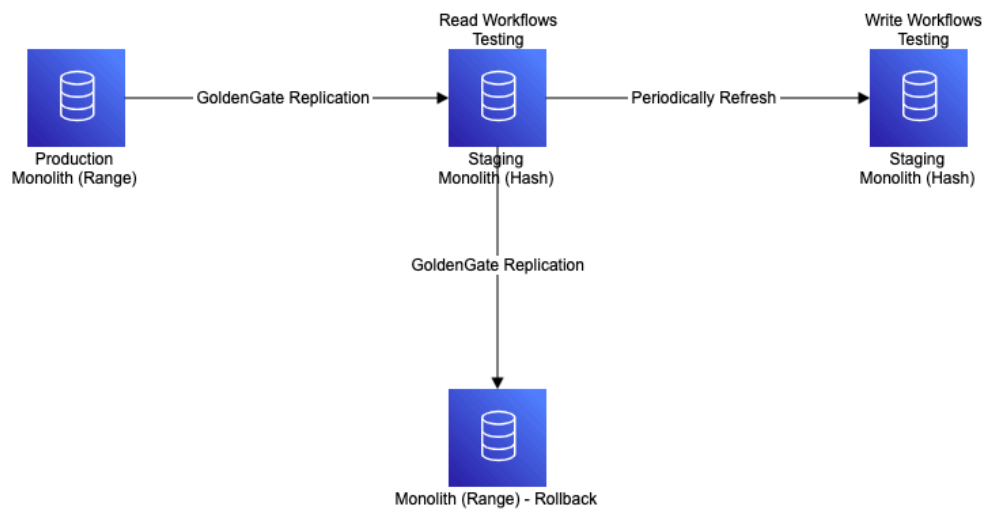
There is opportunity to optimize numeric(10) and numeric(19) data types further. Based on actual values (Min, Max) data analysis, we are following below criteria for these two

- if source datatype is number(10) and current Max value is more than 100M then use BIGINT else use INTEGER
- if source datatype is number(19) and current Max value is more than 100M. Using NUMERIC(19) else use INTEGER
- Use SMALLINT data type in Postgres for NUMBER(1) in Oracle.
- Use INTEGER data type in Postgres for all VERSION columns in all tables to maintain consistency. This has NUMBER(19) data type in Oracle.
- Using SMALLINT for REALM_ID column in all tables. We used for Audit db migration as well.

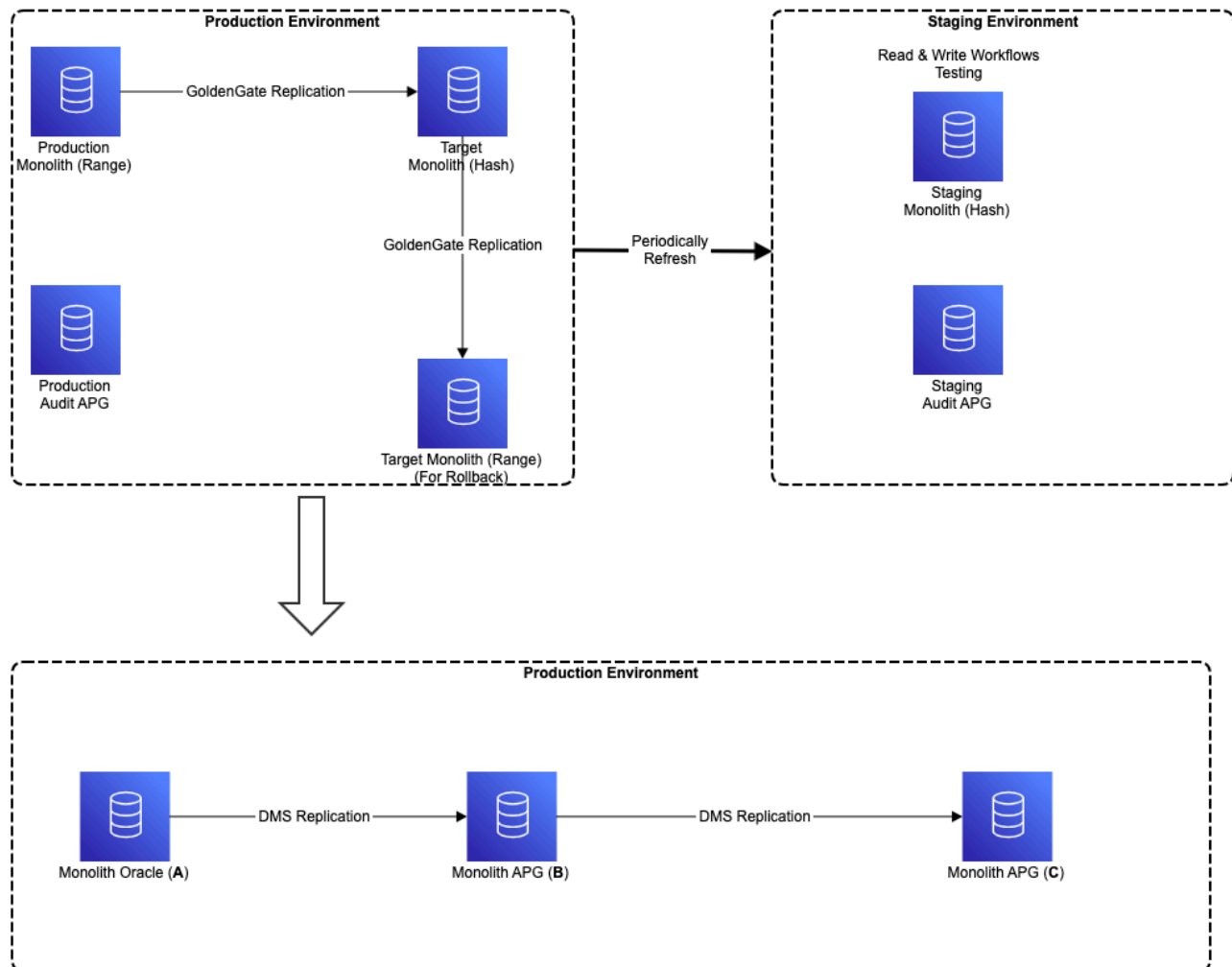
	Oracle	Postgres
1	CLOB	TEXT
2	NUMBER(1)	SMALLINT
3	NUMBER(10)	BIGINT
4	NUMBER(19)	NUMERIC(19,0)
5	NUMBER(19) NOT NULL	NUMERIC(19,0) NOT NULL
6	NUMBER(19,7)	NUMERIC(19,7)
7	TIMESTAMP(6)	timestamp(6) without time zone
8	TIMESTAMP(6) NOT NULL	timestamp(6) without time zone NOT NULL
9	VARCHAR2(100 Char)	character varying(100)
10	VARCHAR2(20 Char)	character varying(20)
11	VARCHAR2(200 Char)	character varying(200)
12	VARCHAR2(255 Char)	character varying(255)

13	VARCHAR2(255 Char) NOT NULL	character varying(255) NOT NULL
14	VARCHAR2(256 Char)	character varying(256)
15	VARCHAR2(30 Char)	character varying(30)
16	VARCHAR2(40 Char)	character varying(40)
17	VARCHAR2(400 Char)	character varying(400)
18	VARCHAR2(4000 Char)	character varying(4000)

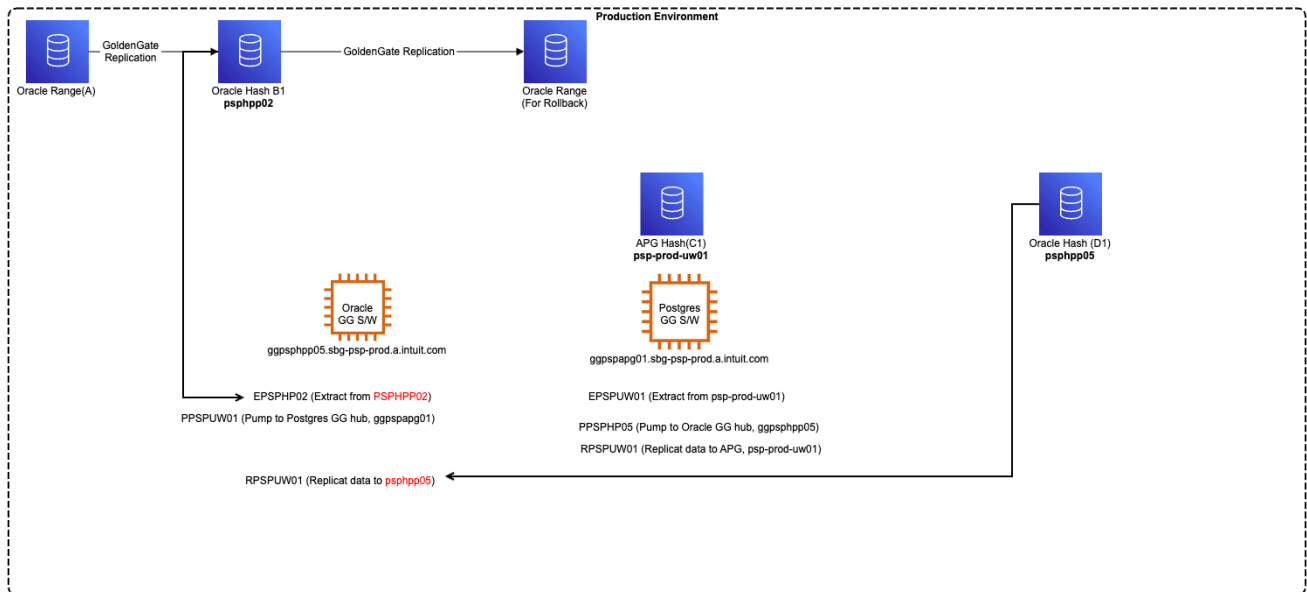
Production Replication and Staging Refresh Strategy



Note: Please note, the Gliffy app will be discontinued. To continue creating and managing diagrams, [Miro](#) is the recommended paved path alternative within Intuit. We encourage you to migrate your diagrams to [Miro](#) to ensure uninterrupted access.



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